Use-Case Flow Specification

Use Case: UC-04 — Identify Idle / Unattached Cloud Assets

Primary Actor: FinOps Analyst

Goal: Identify unattached storage volumes and idle compute instances from ingested AWS, Azure, and GCP billing/usage data and present a monthly savings estimate for each qualifying asset.

Preconditions

• The FinOps Analyst is authenticated and authorized to use the portal.

• Cloud billing CSV data has been successfully ingested and validated for the selected billing period.

• The dataset contains the provider, account/project, asset identifier, cost, and relevant utilization/attachment information required by the detection rules.

Postconditions

• Qualifying unattached storage volumes and idle compute instances are flagged in the optimization results.

• Each flagged asset displays its provider, asset identifier, reason, and estimated monthly savings.

• No asset is modified or deleted by this use case.

Main Success Scenario

1. The FinOps Analyst selects a cloud provider/account/project and billing period for analysis.

2. The system retrieves the validated billing and utilization records for the selected scope.

3. The system evaluates storage records and marks volumes with no active attachment as unattached.

4. The system evaluates compute instances and checks the 14-day utilization window.

5. The system marks a compute instance as idle when its average CPU utilization is below 5%, consistent with FR-001.

6. For each flagged asset, the system calculates an estimated monthly savings amount using the relevant recurring cost data.

7. The system presents the flagged assets, reasons, and savings estimates in the optimization results.

8. The FinOps Analyst reviews the results; the use case ends successfully.

Alternate Flow — 4a. Insufficient or Invalid Utilization Data

• 4a1. The system detects that the selected compute instance does not have sufficient valid utilization data for the required 14-day window.

• 4a2. The system does not classify the instance as idle and labels the record “Insufficient utilization data.”

• 4a3. The system continues evaluating the remaining assets and displays the data-quality status with the optimization results.

• 4a4. The FinOps Analyst may review or correct the source data and rerun the analysis.

Traceability: UC-04 supports FR-002; the compute-idle rule is aligned with FR-001. UC-04 «include» UC-10 (Calculate Monthly Savings Estimate).